# ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** fat-rich food | Bile aids fat digestion. |
| 2 | **(c)** 1,15,200 times | 80 x 60 x 24. |
| 3 | **(d)** to provide a large surface area for absorption | Large area. |
| 4 | **(d)** digestion of all three food types | Pancreatic juice is important. |
| 5 | **(a)** 8000 mL | 16 x 500. |
| 6 | **(a)** plant food takes longer to digest | Cellulose. |
| 7 | **(c)** poor protein digestion and more germs reaching the intestine | HCl needed. |
| 8 | **(c)** 480 litres | 8 L x 60. |
| 9 | **(c)** shorter | Meat is easier to digest. |
| 10 | **(a)** poor absorption of digested food | Less surface area. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Respiration releases energy as heat (1); Metabolism increases after a meal (1). | 1+1 |
| 12 | Right sends blood to lungs nearby (1); Left pumps to the whole body (1). | 1+1 |
| 13 | The air in the alveoli has more oxygen than the blood (1); Oxygen moves into the blood (1). | 1+1 |
| 14 | Muscles need more oxygen (1); Blood must carry it faster (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Fat digestion poor (1); Fewer enzymes (1); Special diet needed (1). | 1+1+1 |
| 16 | Move to site (1); Engulf germs (1); Defence (1). | 1+1+1 |
| 17 | Adds bulk (1); Speeds movement (1); Prevents constipation (1). | 1+1+1 |
| | **OR:** Muscles need more oxygen (1); Heart pumps more (1); Blood removes CO2 faster (1). | 1+1+1 |
| 18 | Insulin is low (1); Blood sugar rises (1); Control diet (1). | 1+1+1 |
| | **OR:** Frog uses oxygen (1); Needs surface air (1); Becomes distressed (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Rest: count breaths per minute (1); After walking (1); After running (1); Table of results (1); Rate rises with activity (1). | 1+1+1+1+1 |
| | **OR:** Lung damage (1); Blood vessel damage (1); Heart disease (1); Toxins strain kidneys (1); Avoid smoking (1). | 1+1+1+1+1 |
| 20 | Heart pumps blood into arteries (1); Arteries carry it to organs (1); Capillaries exchange materials with cells (1); Veins bring blood back (1); Blood returns to the heart (1). | 1+1+1+1+1 |
| | **OR:** Human: lungs, air (1); Fish: gills, water (1); Insect: tracheae (1); Efficiency differs (1); Adaptation to habitat (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 10 million | 5 x 2. |
| 2 | **(c)** the iron-containing haemoglobin | Iron. |
| 3 | **(c)** inhalation is affected | Breathing needs it. |
| 4 | **(d)** 10.5 L | 1.5 x 7. |
| 5 | **(c)** arteries have thick walls to bear high pressure | Pressure. |
| 6 | **(d)** thin air has less oxygen | Compensation. |
| 7 | **(d)** 240 hours | 8 x 30. |
| 8 | **(d)** carbon dioxide | Gas exchange. |
| 9 | **(c)** carbon dioxide has built up | CO2 triggers breathing. |
| 10 | **(d)** 0.83 s | 60/72. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Millions of alveoli (1); Large surface for gas exchange (1). | 1+1 |
| 12 | Adds bulk (1); Prevents constipation (1). | 1+1 |
| 13 | Fats would not digest properly (1); Poor absorption of fatty acids (1). | 1+1 |
| 14 | Dilutes urine (1); Helps pass stones (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Sugar is digested quickly (1); Blood sugar drops (1); Lack of fibre and protein (1). | 1+1+1 |
| 16 | Waves of muscle contraction (1); Moves food along (1); Works in any posture (1). | 1+1+1 |
| 17 | Respiration gives heat (1); Sweat evaporates (1); Blood vessels adjust (1). | 1+1+1 |
|  | **OR:** Dehydration (1); Concentrated wastes (1); Drink more water (1). | 1+1+1 |
| 18 | Training (1); Stronger muscles (1); More oxygen use (1). | 1+1+1 |
|  | **OR:** It is easy to digest (1); It has all the nutrients (1); It protects the baby from infections (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Rice starch to sugar (1); Dal protein to amino acids (1); Vegetables: fibre for egestion (1); Absorbed by villi (1); Used for energy and growth (1). | 1+1+1+1+1 |
|  | **OR:** Filter blood (1); Remove urea (1); Remove extra water as urine (1); Save water when little is drunk (1); Kidney failure is serious (1). | 1+1+1+1+1 |
| 20 | Digestive supplies glucose (1); Respiratory supplies oxygen (1); Circulatory transports both (1); Cells release energy (1); Wastes removed (1). | 1+1+1+1+1 |
|  | **OR:** Press to stop flow (1); Platelets clot (1); Clean wound (1); Cover (1); See a doctor (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** carbon dioxide | Gas exchange in tissues. |
| 2 | **(d)** partial breakdown of glucose without oxygen forms lactic acid | Anaerobic respiration. |
| 3 | **(d)** beats less often but pumps more per beat | Efficient. |
| 4 | **(d)** kidney | Made from proteins. |
| 5 | **(d)** heart, aorta, arteries, capillaries, veins, heart | Correct circuit. |
| 6 | **(a)** aorta, arteries, capillaries | Decreasing size. |
| 7 | **(a)** water, urea and salts | Waste. |
| 8 | **(a)** urea | Artificial kidney. |
| 9 | **(c)** 6 to 7 m | Long coiled tube. |
| 10 | **(d)** teeth | Teeth are mechanical. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Damages the lining of the food pipe and stomach (1); Causes acidity (1). | 1+1 |
| 12 | Germs and dirt are removed (1); Stomach infections are avoided (1). | 1+1 |
| 13 | Cells need constant supply (1); Wastes must be removed (1). | 1+1 |
| 14 | Aerobic uses oxygen (1); Anaerobic occurs without oxygen (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | More oxygen needed (1); Carbon dioxide removed (1); Larger volume per breath (1). | 1+1+1 |
| 16 | It is the number of throbs of an artery per minute (1); It equals the heartbeat rate (1); It rises with exercise (1). | 1+1+1 |
| 17 | Less oxygen carrying (1); Muscles need more (1); Breathing speeds up (1). | 1+1+1 |
| | **OR:** Stomach: acid and pepsin act on proteins (1); Small intestine: bile and pancreatic juice act on all foods (1); Digestion is completed in the small intestine (1). | 1+1+1 |
| 18 | One-cell thick wall (1); Easy exchange (1); Close to every cell (1). | 1+1+1 |
| | **OR:** Germs (1); Cause diarrhoea (1); Contaminated water (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Herbivore flat teeth (1); Herbivore long intestine (1); Carnivore sharp teeth (1); Short intestine (1); Different digestion (1). | 1+1+1+1+1 |
| | **OR:** Acidity: spicy food (1); Ulcers: acid damage (1); Constipation: low fibre (1); Diarrhoea: germs (1); Hygiene and diet (1). | 1+1+1+1+1 |
| 20 | Tiredness (1); Infections (1); Iron-rich diet (1); Medical advice (1); Rest (1). | 1+1+1+1+1 |
| | **OR:** Food has chemical energy (1); Digestion gives glucose (1); Respiration releases energy (1); Used for movement (1); Excess stored as fat (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.